The Power Rule (template) — teacher key

Question: How fast does x? change, for any n?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	4	never trimmed
predict	1	4	never trimmed
explore	3	8	skipped first
build	2	8	shrinks to 2 of 4
practice	1	8	never trimmed
check	1	4	never trimmed
Huddle h1	clock	minute 18	pulled in by the clock

Targets

- I can state the power rule in words.
- I can differentiate a polynomial term by term.
- I can find the slope at a point.

Predictions

- The derivative of x^2 is $2x$. What do you expect for x^3? $3x^2$. Bring the 3 down and lower the power: $3x^2$.
- What is the slope of the flat line $y = 7$? **0.** A flat line does not rise: the derivative is 0.

Huddle script (answers in green)

Definitions

- **Derivative:** The **slope** of the tangent line at a point.
- **Power rule:** Bring the **power** down, then **subtract 1** from it.

Worked example

Differentiate f(x) = $4x^3$.

- Bring down the power: 3 · 4x
- Lower the power by 1: 12x**2**
- So f?(x) = **$12x^2$**

Label the slope

1. The **point** where we measure the slope.
2. The **tangent line**; its slope is f?.

Every level, with answers

Differentiate

Level 1. Differentiate.
Answer: $4x^3$ Bring down 4 and lower the power: $4x^3$.

Level 2. Differentiate.
Answer: $12x$ Multiply 6 by 2 to get 12, then lower the power: $12x$.

Level 3. Differentiate term by term.
Answer: $10x^4 + 5$ $10x^4 + 5 + 0$. The constant 1 disappears.

Level 4. Differentiate. Rewrite the root as a power if that helps.
Answer: $\frac{6}{2\sqrt{x}}$ $6x^{1/2}$ becomes $\frac{6}{2}x^{-1/2} = \dfrac{6}{2\sqrt{x}}$.

Slope at a point

Level 1. Let $f(x) = x^2$. Find the slope of the curve at $x = 2$.
Answer: 4 $f'(x) = 2x$, so $f'(2) = 4$.

Level 2. Let $f(x) = 4x^2$. Find the slope of the curve at $x = 3$.
Answer: 24 $f'(x) = 8x$, so $f'(3) = 24$.

Level 3. Let $f(x) = 2x^4$. Find the slope of the curve at $x = 3$.
Answer: 216 $f'(x) = 8x^3$, so $f'(3) = 216$.

Level 4. Let $f(x) = 4x^4$. Find the slope of the curve at $x = 1$.
Answer: 16 $f'(x) = 16x^3$, so $f'(1) = 16$.

Common misconceptions

Students say	Say instead
Differentiate x^3 and get $3x^3$.	The power drops by 1 as well as coming down.
The derivative of 7 is 7.	A constant does not change, so its slope is 0.
Differentiate $5x^2$ and get $5 \cdot 2x \cdot \ldots$	Multiply the 5 by the power once: 10x.